Deep learning, TinyML and Open source

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PLAN

- Lensless imaging
- Reconstruction methods
- Goals and motivations
- Model deployment
- Results
- Summary
- Next

Lensless imaging : Context

- Pros
  - High-resolution
  - Focalization
- Cons
  - Expensive
  - Weight
  - Space

What if remove lens ?

RECONSTRUCT

LENSLESS IMAGING: IDEA

mask sensor

RECONSTRUCTION METHOD

BENEFITS

- ✓ Low-cost, light-weight, space
- ✓ Miniaturization

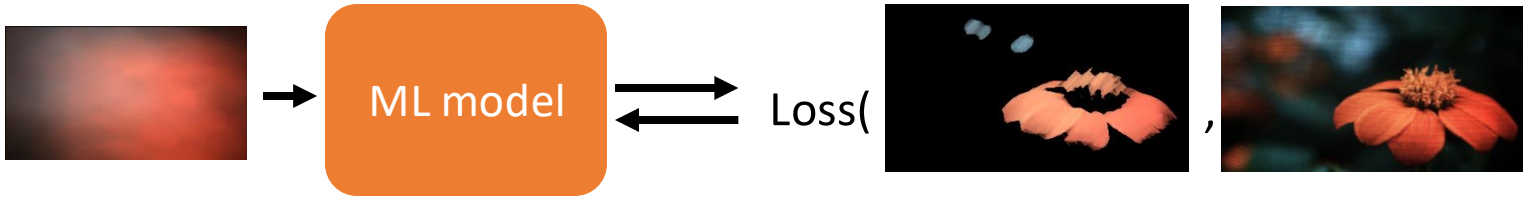
APPLICATIONS

- ➤ Medical systems
- ➤ Wearables
- ➤ IoT devices

RECONSTRUCTION METHODS

1. CLASSICAL ALGORITHMS
   - Based signal processing theory
   - Iterative algorithms
   - FISTA, ADMM, gradient descent

2. DEEP LEARNING
   - Need dataset



3. UNROLLED OPTIMIZATION
   - Mix classic + deep learning

RECONSTRUCTION: PROJECT TARGET

- Study two deep learning approaches
  - Monakhova et al. (Learned reconstruction) + Khan et al. (FlatNet)

- Benefits of ML

Ground truth

ADMM

Unrolled ADMM

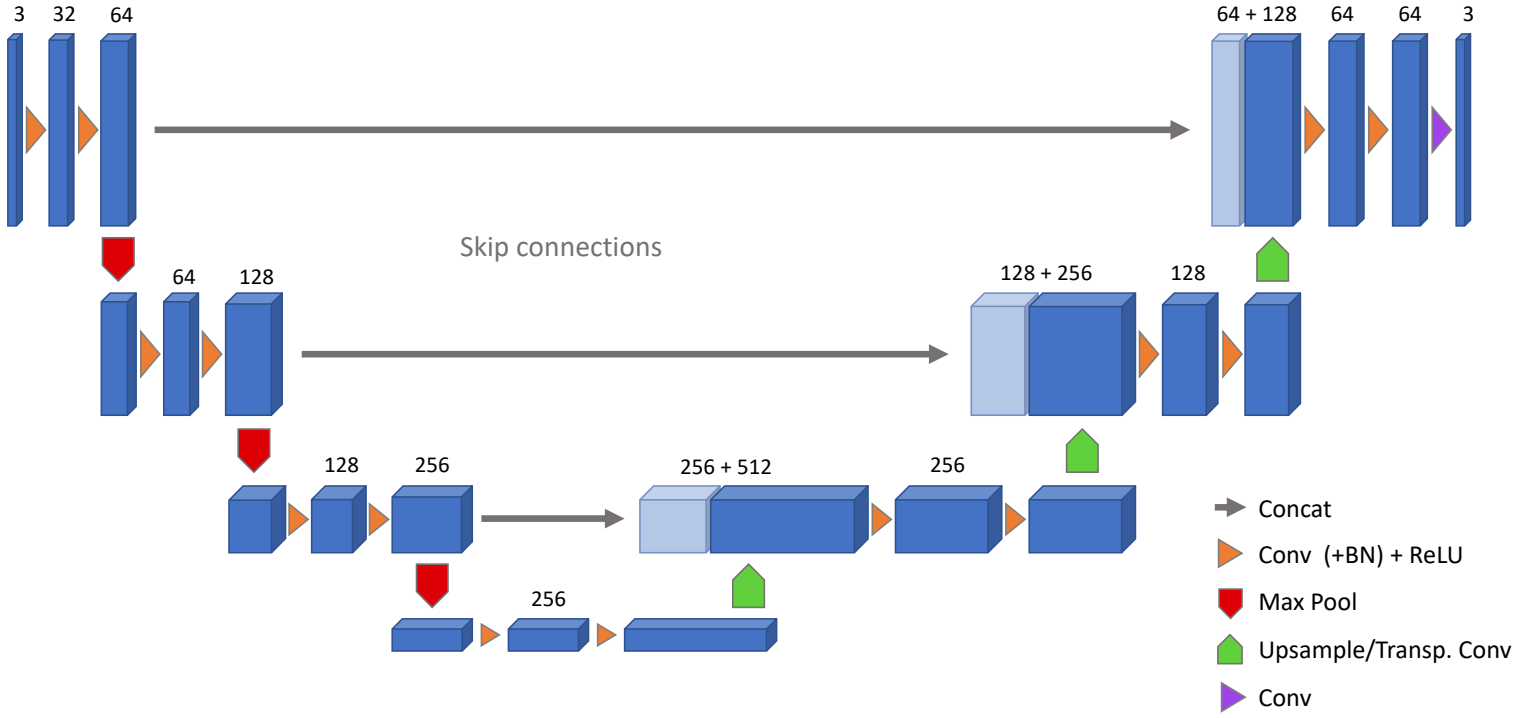
ML

Running time on GPU (ms)	1’520	71	10

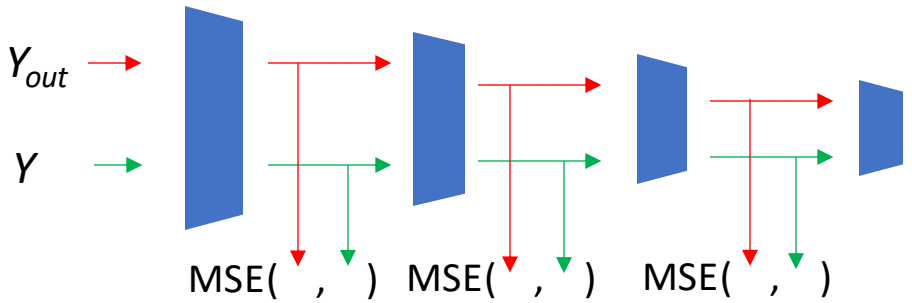
MONAKHOVA ET AL.

- Based on U-Net

- Loss
  - Mean-squared error (MSE)
    - $Loss(Y_{out}, Y) = \|Y_{out} - Y\|_2^2$
    - Pixelwise comparison

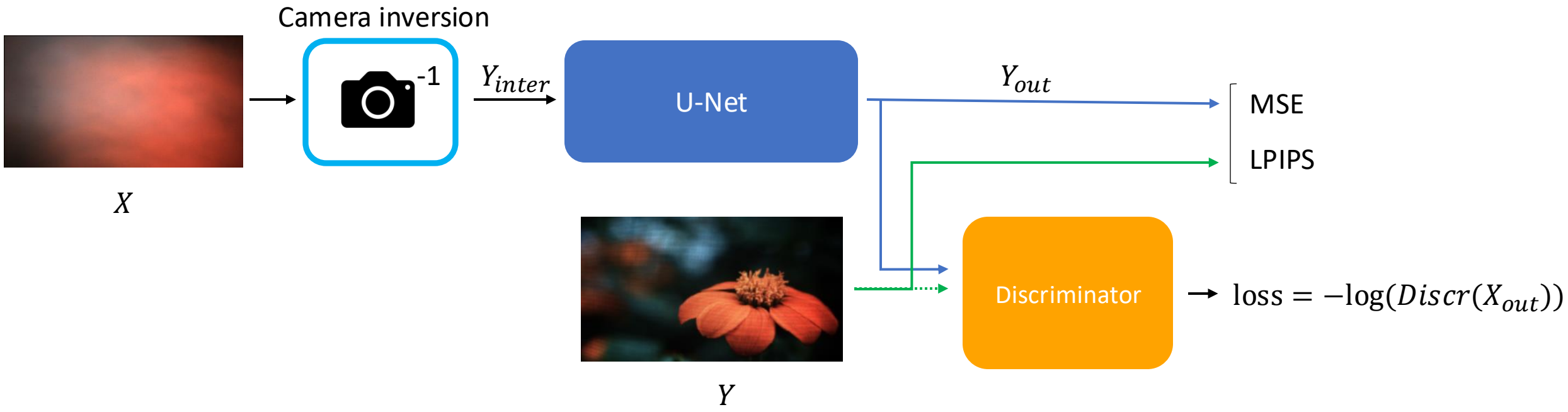


  - LPIPS error
    - Use pretrained model
    - Different models : AlexNet, VGG
    - Capture perception : edges, shapes



"Learned reconstructions for practical mask-based lensless imaging" [2019, K. Monakhova]

KHAN ET AL.: OVERVIEW



"FlatNet: Towards Photorealistic Scene Reconstruction from Lensless Measurements" [2020, S. Khan]

KHAN ET AL.: SETTING-UP

- Inverse problem formulation

$$X = H \cdot Y$$

- Invert H effect → H^{-1}
- H^{-1} undetermined

→ need simplification : **with mask**

Khan et al.: Setting-up

Separable mask

- Inner-product of 2 vectors

$$X = \underbrace{\Phi_L}\, Y\, \underbrace{\Phi_R^T}$$

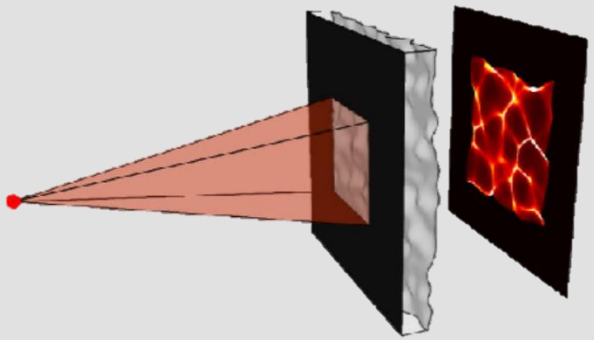
Linear operators (matrix form)

Non-separable mask

- Assume PSF shift-invariant

$$X = Y * \underbrace{h}$$

PSF of camera



KHAN ET AL.: CAMERA INVERSION

Idea : invert operation → learnable layer

SEPARABLE MASK

$$X = \Phi_L Y \Phi_R^T$$

$$Y_{inter} = f(W_1 \; X \; W_2)$$

Φ_R^T adjoint

Φ_L adjoint

Non-linearity: leaky ReLU

NON- SEPARABLE MASK

$$X = Y * h$$

$$\mathcal{F}(X) = \mathcal{F}(Y) \odot H$$

Pointwise multiplication

$$Y_{inter} = \mathcal{F}^{-1}(\mathcal{F}(W) \odot \mathcal{F}(X))$$

Master thesis: subject

Goal

- Implement Monakhova et al. + Khan et al. models
- Deploy to embedded systems

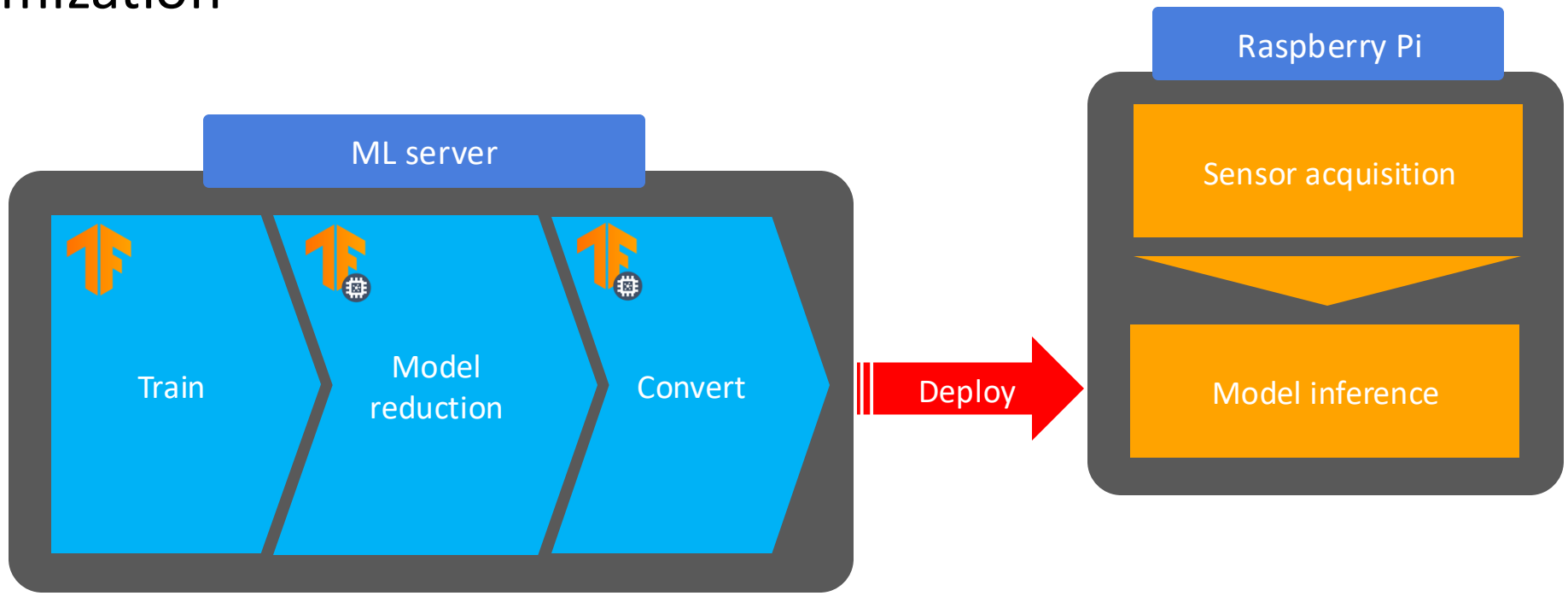
Motivations

- Reproducibility + availability
- Usable in various devices : phones, Raspberry Pi

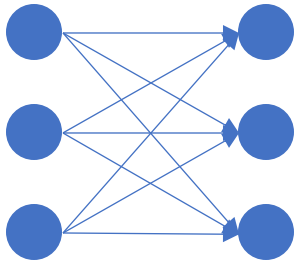
WORKFLOW

- Need reduce size model for embedded systems
  - Flash : model parameters + architecture
  - RAM : computation, store temporary results

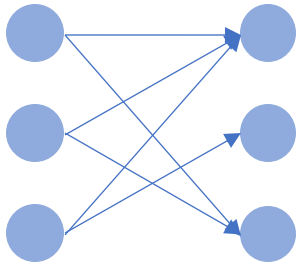
→ Model optimization



Model optimization: Weight pruning



3.32	29.01	27.07
0.9	50.3	-21
8.9	-0.1	7.9



3.32	0	27.07
0.9	0	-21
8.9	-0.1	0

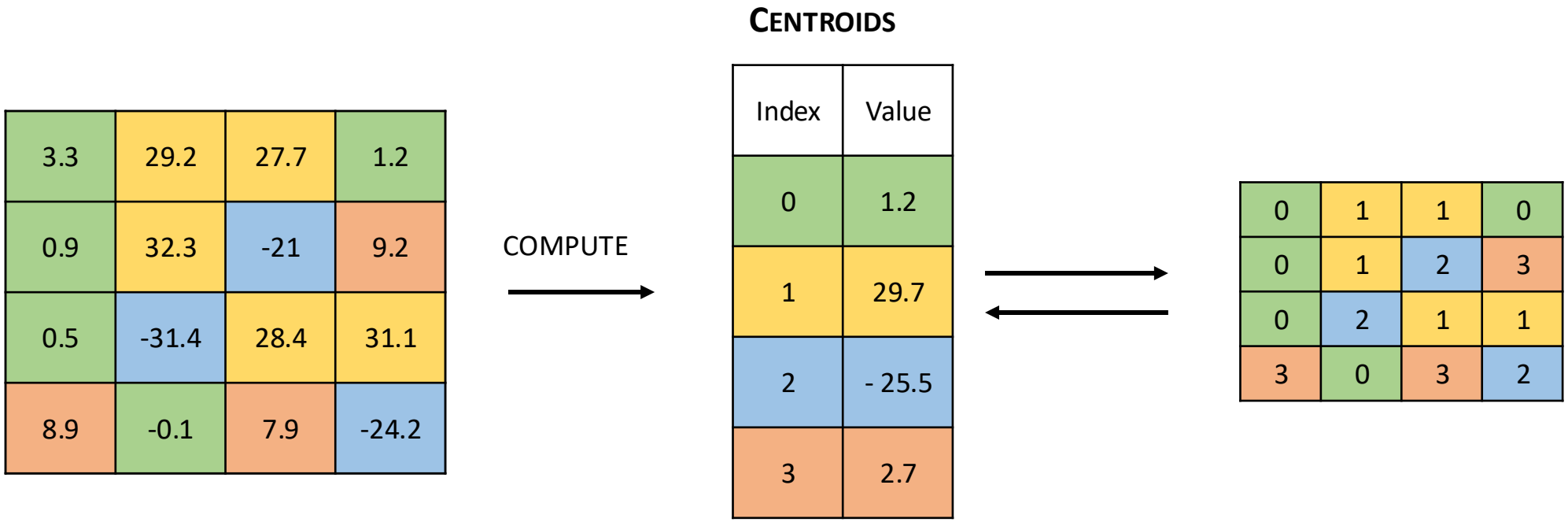
- Magnitude pruning

$$w_i = \begin{cases} w_i & if \ |w_i| > \lambda \\ 0 & otherwise \end{cases}$$

- Sensivity pruning

  λ dependent to layer's std. deviation

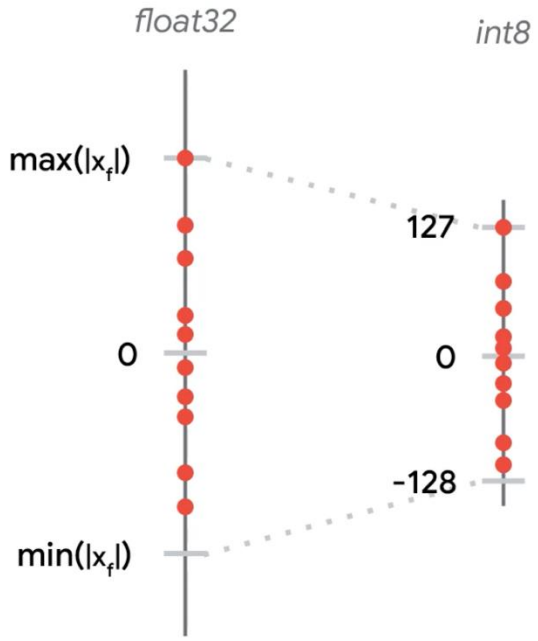
Model optimization: Clustering



MODEL OPTIMIZATION: QUANTIZATION

Idea : floating point : 32 bits (float32) → integer : 8 bits (int8)

- Weight quantization:
  - Compress weights int8
  - Decompress + compute in float32
  - ~4x reduction Flash
- Inference quantization
  - Store + compute in int8
  - Quantization-aware training (QAT)
  - Reduction Flash + RAM

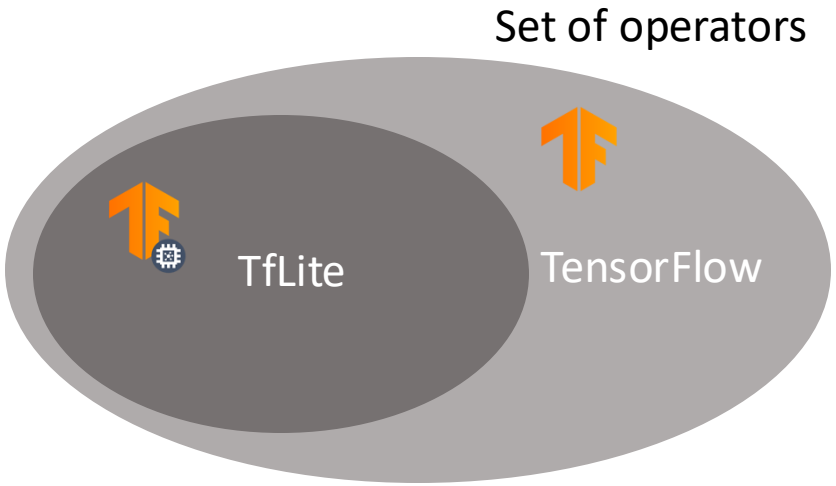


Model optimization: TensorFlow Lite

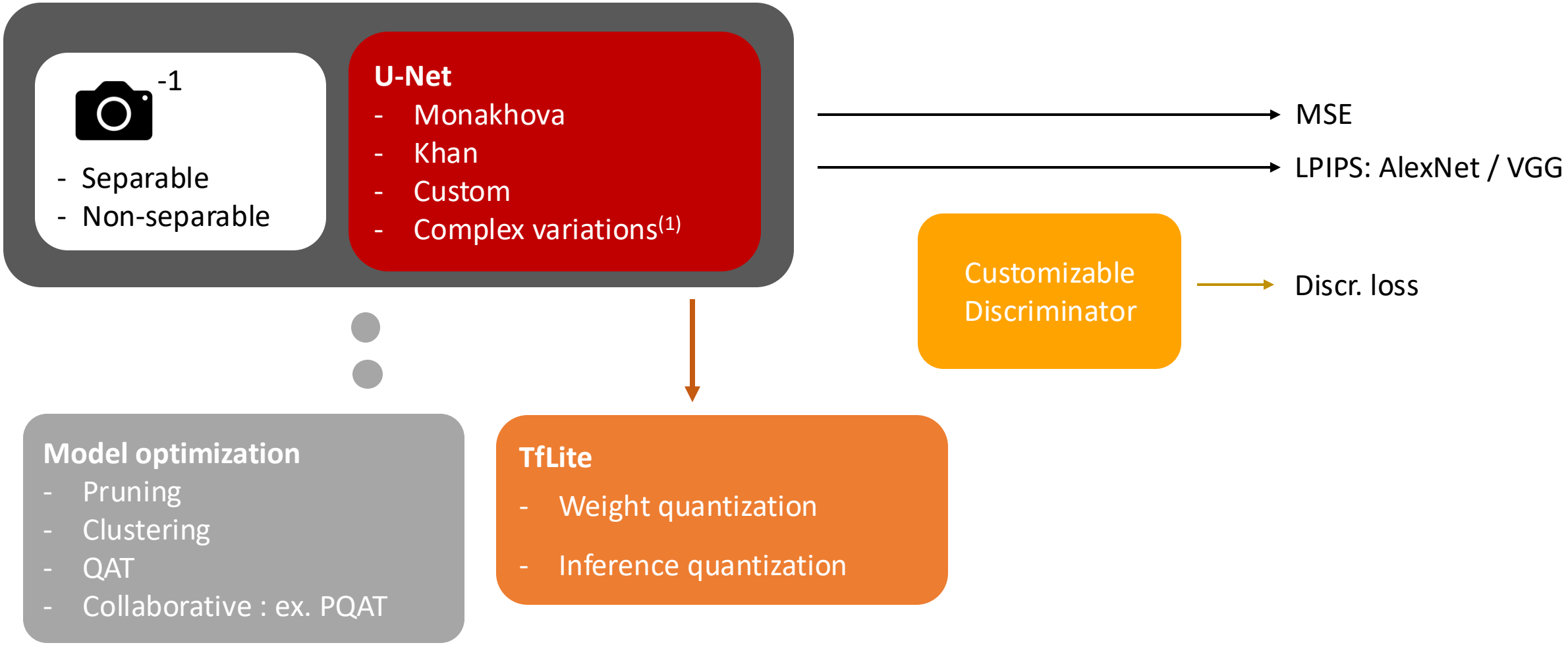
TensorFlow not for embedded systems

TfLite format : for Python, C++ apps

- Not hardware dependent
- No external tools needed



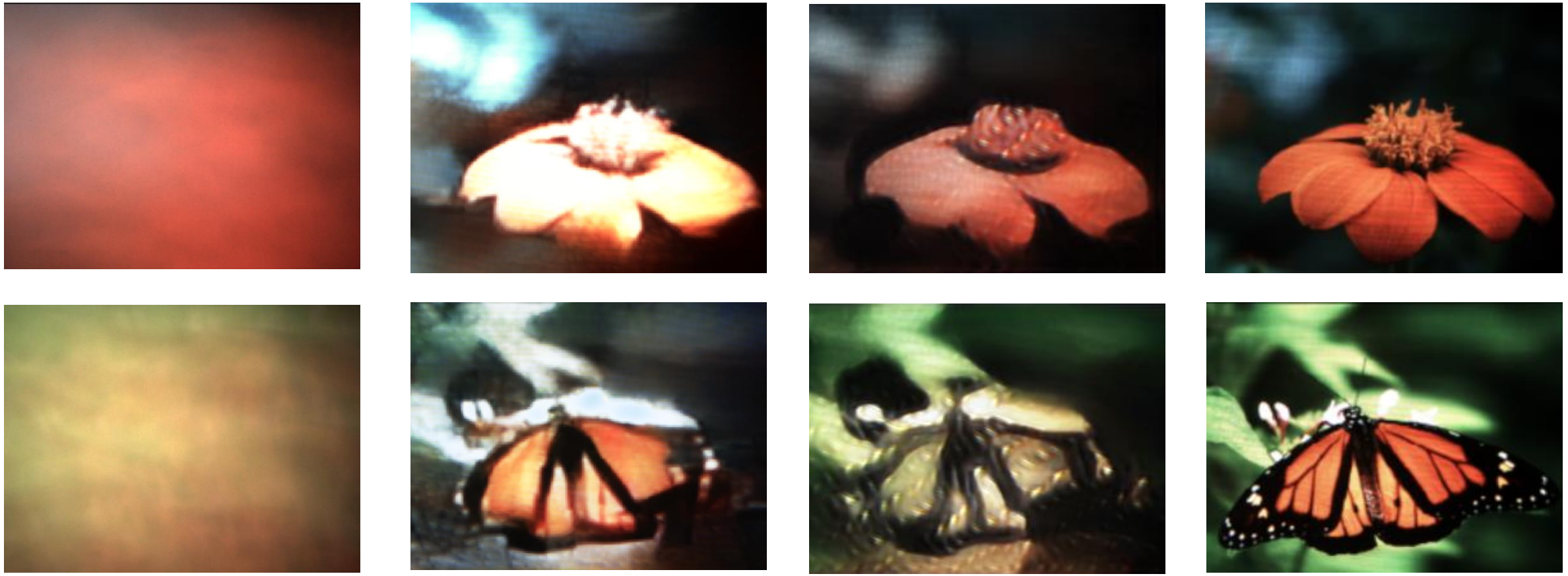
Implementation: Overview



(1) "*Keras-unet-collection*" [2021, Y. Sha]

RESULTS: MONAKHOVA ET AL.

- DiffuserCam dataset
  - 25’000 RGB images
  - Shift-invariant PSF (non-separable)



Raw measurements | Expected | Ours | Ground truth

	Expected	Ours
LPIPS (lower ✓)	0.246	**0.231**
MSE (lower ✓)	**0.015**	0.038

Results: Khan et al.

Ground truth

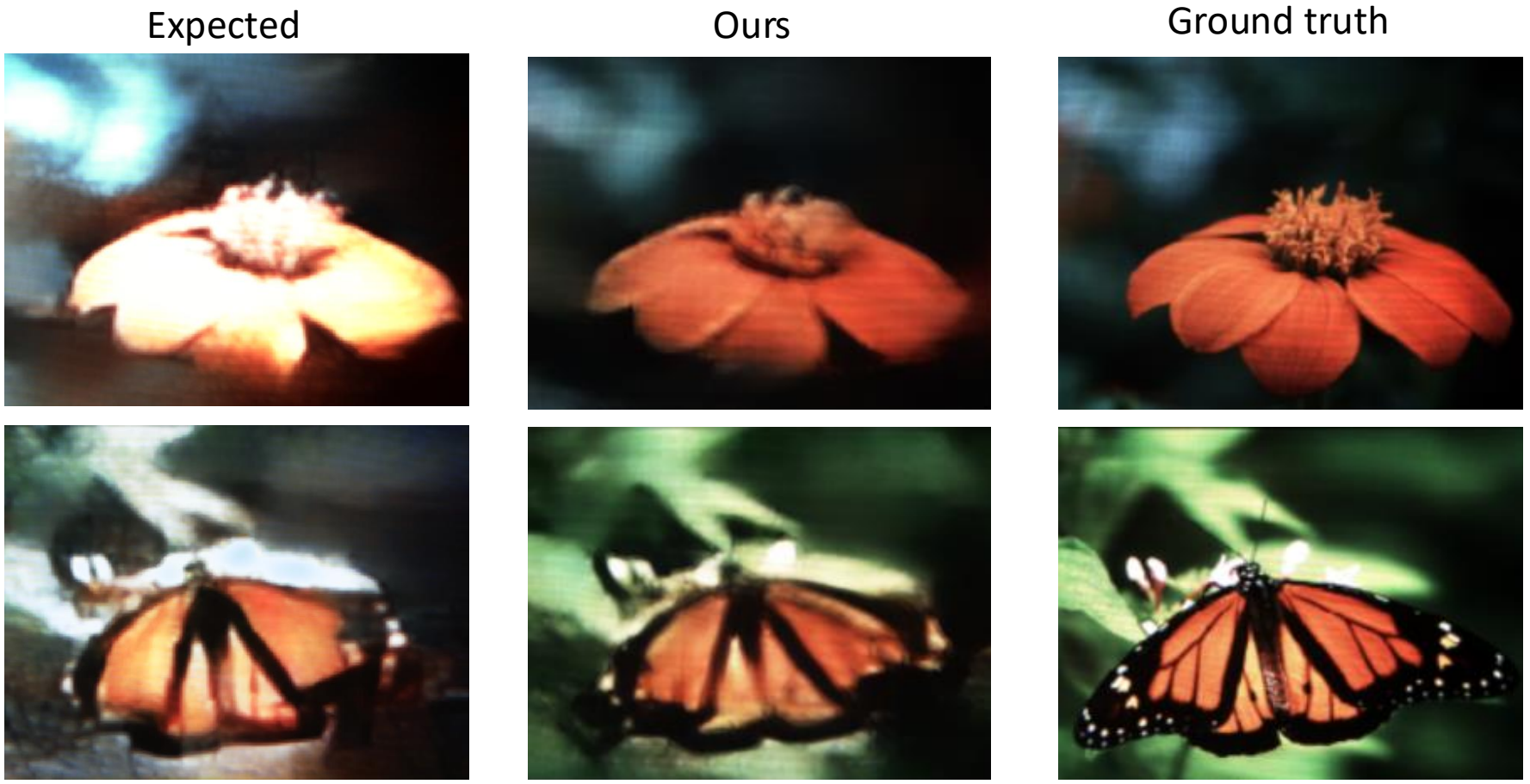
Ours

- FlatCam dataset
  - 10’000 images
  - Bayer measurements
  - Separable mask

	Expected	Ours
PSNR (higher ✓)	19.62	**19.80**
SSIM (higher ✓)	**0.64**	0.53

RESULTS: FLATNET

DiffuserCam (non-separable) + inversion layer



Results: Model reduction

Ground truth

Normal

Weight quantization

Pruned

Inference quantization

Results: Model reduction

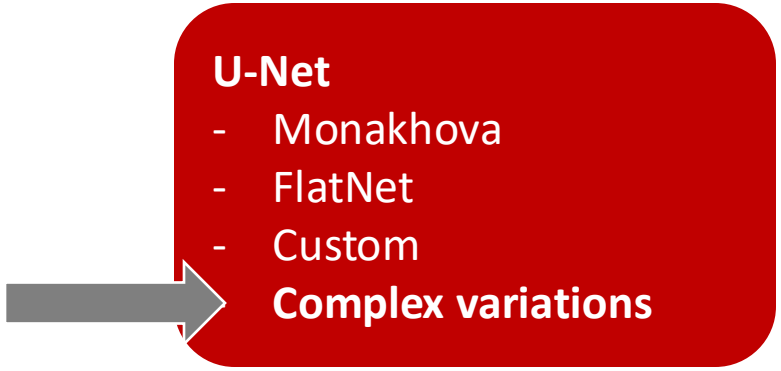
	Normal	Pruned **+ quant**	QAT
LPIPS	**0.38**	0.41	0.46
MSE	**0.039**	0.045	0.095
Flash (zip, MB)	40.1	25 → **7**	**9.4**
RAM (MB)	320	320	**130**
Time (1 core, s)	1.74	1.56	**1.39**

SUMMARY

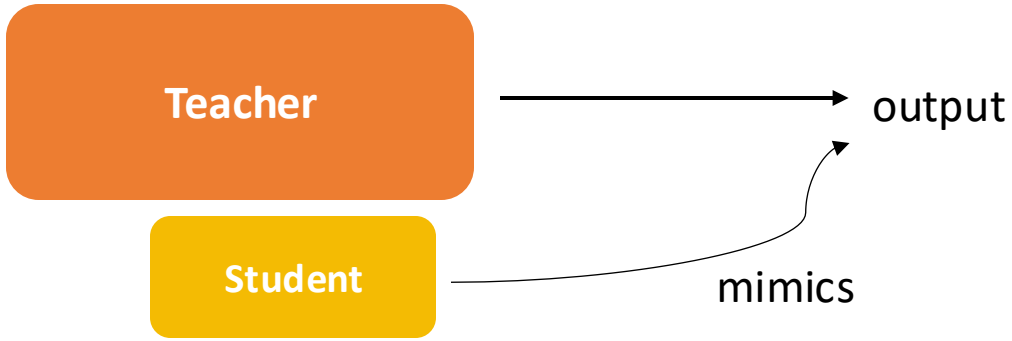
- Study Monakhova et al. + Khan et al.
- Deployment
  - Model optimization
  - TensorFlow Lite
- Importance of inversion layer
- Effects of model optimization

NEXT

- Test more complex models



- Knowledge distillation

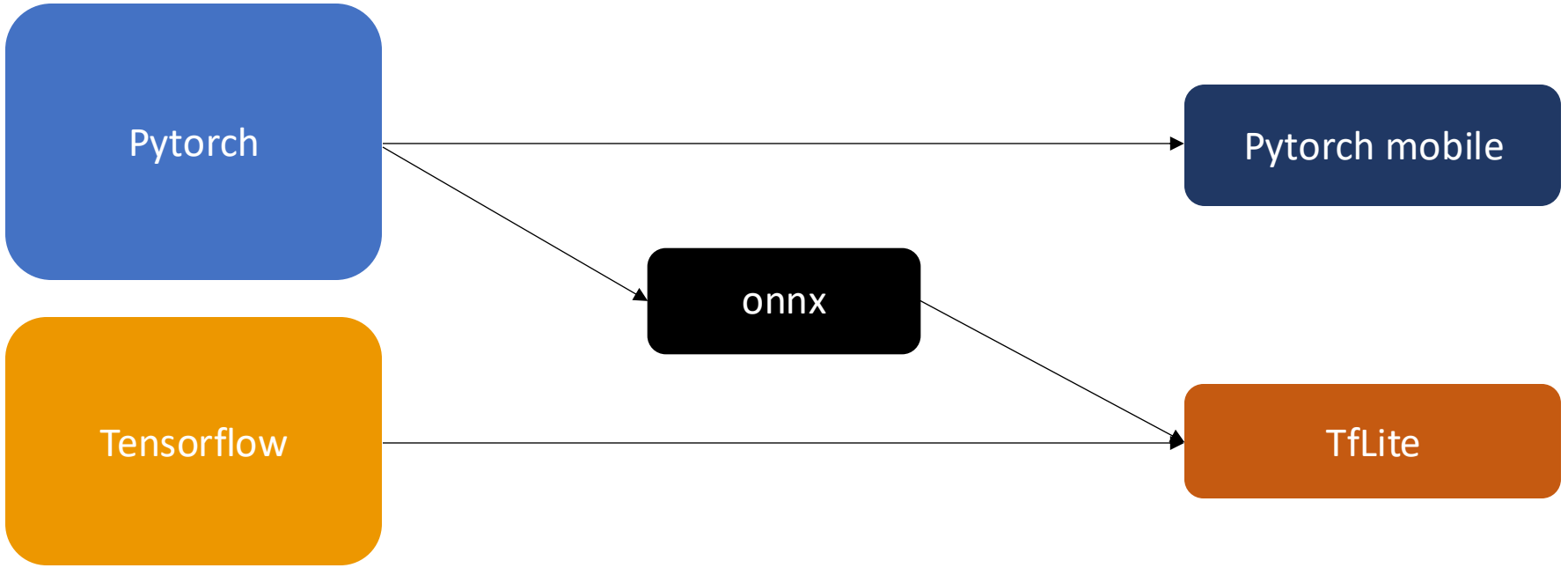


- Deploy image classification models

THANK YOU

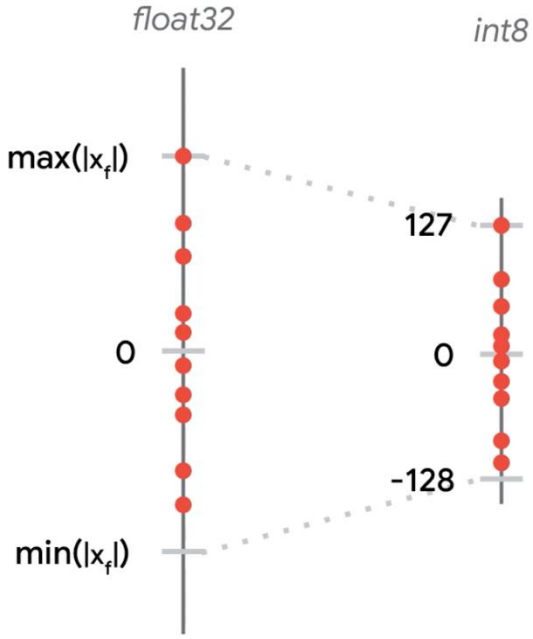
for your attention

WHY TENSORFLOW?



- Less model optimization techniques (ex. clustering)
- Unknown operators during conversion

MODEL OPTIMIZATION: QUANTIZATION



- Idea : floating point : 32 bits (float32) → integer : 8 bits (int8)
- Weight quantization:
  - Compress weights int8:

$$x_{int} = \frac{x_{float} - min_f}{max_f - min_f} * 255 \; - 128$$

  - Decompress + compute in float32:

$$x_{float} = \frac{x_{int} - min_f}{255} * (max_f - min_f) -, min_f$$

  - ~4x reduction Flash
- Inference quantization
  - Store + compute in int8
  - Can be done in training : Quantization-aware training (QAT)
  - ~4x reduction Flash + RAM